

AskaryanPaper4

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1 The Derivation

Let $\epsilon = \omega_0/\omega_C$. The signal model is:

$$s(t) = E_0 \begin{cases} (1 - \frac{1}{2}\epsilon) e^{\omega_0 t} & \text{if } t < 0 \\ 2e^{-2\omega_C t} - (1 + \frac{1}{2}\epsilon) e^{-\omega_0 t} & \text{if } t \geq 0 \end{cases} \quad (1)$$

Let $u(t)$ represent the Heaviside step function. The response model is:

$$r(t) = R_0 \cos(2\pi f_0 t) e^{-2\pi\gamma t} u(t) \quad (2)$$

The task is to convolve $r(t)$ and $s(t)$. For the case $t < 0$, the convolution is

$$s * r = E_0 R_0 \Re \int_{-\infty}^{\infty} \left(1 - \frac{1}{2}\epsilon\right) e^{\omega_0 t} e^{-\omega_0 \tau} e^{2\pi j f_0 \tau} e^{-2\pi\gamma \tau} u(\tau) d\tau \quad (3)$$

Factor terms that do not involve τ , and account for the step function:

$$s * r = E_0 R_0 \left(1 - \frac{1}{2}\epsilon\right) e^{\omega_0 t} \Re \int_0^{\infty} e^{-\omega_0 \tau} e^{2\pi j f_0 \tau} e^{-2\pi\gamma \tau} d\tau \quad (4)$$

Let $k = \omega_0 + 2\pi\gamma - 2\pi j f_0$. We find

$$s * r = E_0 R_0 \left(1 - \frac{1}{2}\epsilon\right) e^{\omega_0 t} \Re \int_0^{\infty} e^{-k\tau} d\tau \quad (5)$$

The final result for $t < 0$ is

$$\boxed{s * r = E_0 R_0 \left(1 - \frac{1}{2}\epsilon\right) \Re(k^{-1}) e^{\omega_0 t}} \quad (6)$$

The result is a rising exponential ($t < 0$) that does not oscillate at the response frequency f_0 . Let r_a be the analytic signal for $r(t)$, and let $s_+(t)$ and $s_-(t)$ be the piecewise branches of $s(t)$. For the case of $t \geq 0$, we have

$$s * r = E_0 R_0 \Re \left(\int_{-\infty}^t s_+(t - \tau) r_a(\tau) d\tau + \int_t^{\infty} s_-(t - \tau) r_a(\tau) d\tau \right) \quad (7)$$

Using Eqs. 1 and 2, we find

$$s * r = E_0 R_0 \Re \left(2e^{-2\omega_C t} \int_0^t e^{2\omega_C \tau} r_a(\tau) d\tau - \left(1 + \frac{1}{2}\epsilon\right) e^{-\omega_0 t} \int_0^t e^{\omega_0 \tau} r_a(\tau) d\tau + \left(1 - \frac{1}{2}\epsilon\right) e^{\omega_0 t} \int_t^{\infty} e^{-\omega_0 \tau} r_a(\tau) d\tau \right) \quad (8)$$

Define the following constants, with units of frequency:

$$z_1 = 2\omega_C + 2\pi j f_0 - 2\pi\gamma \quad (9)$$

$$z_2 = \omega_0 + 2\pi j f_0 - 2\pi\gamma \quad (10)$$

$$z_3 = -\omega_0 + 2\pi j f_0 - 2\pi\gamma \quad (11)$$

Using these definitions and Eq. 2, the convolution becomes

$$s * r = E_0 R_0 \Re \left(2e^{-2\omega_C t} \int_0^t e^{z_1 \tau} d\tau - \left(1 + \frac{1}{2}\epsilon\right) e^{-\omega_0 t} \int_0^t e^{z_2 \tau} d\tau + \left(1 - \frac{1}{2}\epsilon\right) e^{\omega_0 t} \int_t^{\infty} e^{z_3 \tau} d\tau \right) \quad (12)$$

Evaluating the integrals and distributing the $\Re$ operator:

$$s * r = E_0 R_0 \left(2e^{-2\omega_C t} \Re(z_1^{-1} e^{z_1 t} - z_1^{-1}) - \left(1 + \frac{1}{2}\epsilon\right) e^{-\omega_0 t} \Re(z_2^{-1} e^{z_2 t} - z_2^{-1}) - \left(1 - \frac{1}{2}\epsilon\right) e^{\omega_0 t} \Re(z_3^{-1} e^{z_3 t}) \right) \quad (13)$$

The real part of the inverse of a complex number z is

$$\Re(z^{-1}) = z^{-2} \Re(z) \quad (14)$$

Let $z = x + jy$. The real part of a complex exponential divided by z is

$$\Re(z^{-1} e^{zt}) = z^{-2} e^{xt} (x \cos(yt) + y \sin(yt)) \quad (15)$$